

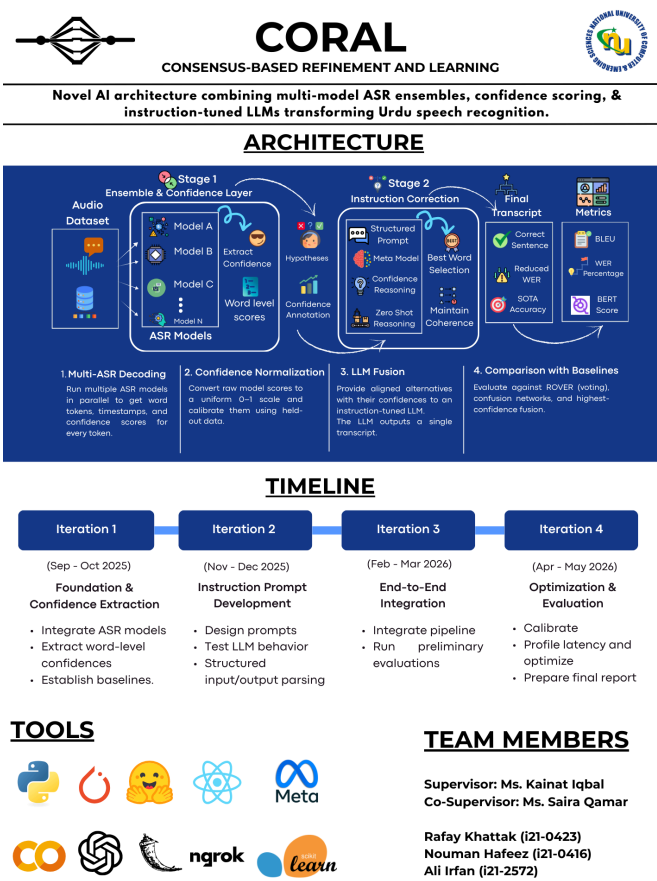
CORAL - Consensus-based Refinement And Learning for Urdu ASR

CORAL is a novel AI architecture designed to overcome the performance ceiling of low-resource Urdu Automatic Speech Recognition (ASR). Since traditional models struggle with domain brittleness and error propagation, CORAL solves this via a two-stage machine learning pipeline. First, an ensemble of acoustic models (Whisper, Wav2Vec2) generates multiple transcript hypotheses alongside word-level confidence scores. Second, an instruction-tuned Large Language Model (LLM) intelligently synthesizes these confidence-annotated hypotheses to produce a highly accurate, error-corrected final transcript without requiring expensive acoustic fine-tuning.

Features include:

- 1) Multi-model hypothesis generation utilizing diverse ASR architectures.
- 2) Few-shot (In-Context Learning) LLM-guided post-processing for complex phonetic and grammatical error correction.
- 3) Word-level confidence extraction to precisely guide the LLM's consensus reasoning.
- 4) Interactive web interface for real-time audio processing and model output comparison.

Key Words: Urdu ASR, Large Language Models, Ensemble Learning, Confidence Estimation, Speech-to-Text



Technology Used:

AI & LLMs:

Alif-1.0-8B, Qalb-8B, HuggingFace Transformers, BitsAndBytes (4-bit Quantization)

Acoustic ASR Models:

Whisper v3 (OpenAI), Wav2Vec2-XLSR

Backend & Processing:

Python, PyTorch, Flask REST API, Librosa

Web (Frontend):

Next.js, React, Tailwind CSS

Infrastructure:

Google Colab (GPU Inference), Ngrok

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